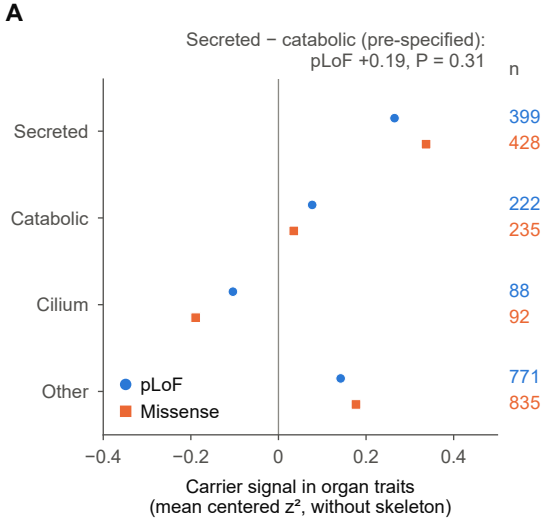

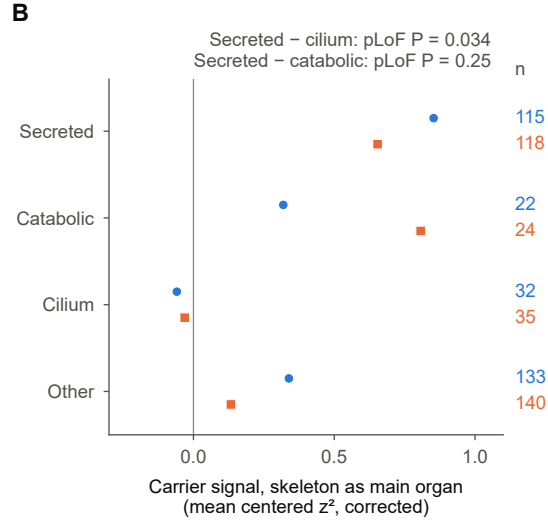

C

	pLoF			Missense		
	Primary	Rank test	Without top 25	Primary	Rank test	Without top 25
W1 GO main product	+0.82 P = 0.008	+0.043 P = 0.059	+0.54 P = 0.024	+1.25 P = 0.004	+0.017 P = 0.286	+0.84 P = 0.013
W2a expressed in organ	+0.12 P = 0.121	+0.028 P = 0.025	+0.12 P = 0.041	+0.22 P = 0.025	+0.024 P = 0.041	+0.17 P = 0.035
W2b organ-specific	+0.26 P = 0.007	+0.029 P = 0.020	+0.16 P = 0.010	+0.47 P = 0.0001	+0.049 P = 0.0003	+0.32 P = 0.0003
W3 W1 and W2a	+0.65 P = 0.088	+0.015 P = 0.373	+0.49 P = 0.083	+1.55 P = 0.007	+0.022 P = 0.287	+1.01 P = 0.020

Rule routes outside the skeleton: difference rule genes − other genes
(shading, $-\log_{10}$ P; α = 0.0125)

D

	Carriers, rule / other genes	s_{het}, rule / other genes	Adjusted effect (P)
W1, pLoF	102 / 148	0.0118 / 0.0080	+0.58 (P = 0.040)
W2b, pLoF	108 / 162	0.0119 / 0.0068	+0.23 (P = 0.025)
W1, Missense	26,704 / 43,282	0.0120 / 0.0081	+0.99 (P = 0.025)
W2b, Missense	26,704 / 55,376	0.0122 / 0.0069	+0.49 (P = 0.0002)